

Homework 6 Solution

1. For each of the following matrices, find an invertible matrix P for which $P^{-1}AP$ is diagonal, or prove that it doesn't exist.

(a)

$$A = \begin{pmatrix} i & i \\ i & i \end{pmatrix}$$

Solution:

The characteristic polynomial of A is given by

$$p_A(\lambda) = \det(\lambda I_2 - A) = \det \begin{pmatrix} \lambda - i & -i \\ -i & \lambda - i \end{pmatrix} = (\lambda - i)^2 - i^2 = \lambda(\lambda - 2i)$$

Equating to zero, the eigenvalues are given by $\lambda_1 = 0, \lambda_2 = 2i$. We now find a basis for each of the eigenspaces

$$V_{\lambda_1} = \text{Null}(0 \cdot I_n - A) = \text{Null} \begin{pmatrix} -i & -i \\ -i & -i \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

Thus, a basis for V_{λ_1} is the vector $\vec{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$V_{\lambda_2} = \text{Null}(2i \cdot I_2 - A) = \text{Null} \begin{pmatrix} i & -i \\ -i & i \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

Thus, a basis for V_{λ_2} is the vector $\vec{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

As $\vec{v}_1, \vec{v}_2$ are a basis of $\mathbb{R}^2$, consisting of eigenvectors of A , then A is diagonalizable, $P = \begin{pmatrix} \vec{v}_1 & \vec{v}_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ and

$$P^{-1}AP = D = \begin{pmatrix} 0 & 0 \\ 0 & 2i \end{pmatrix}.$$

(b)

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

Solution: The characteristic polynomial of A is given by

$$p_A(\lambda) = \det(\lambda I_2 - A) = \det \begin{pmatrix} \lambda - 2 & 0 \\ -1 & \lambda - 2 \end{pmatrix} = (\lambda - 2)^2$$

Equating to zero, the eigenvalues are given by $\lambda_{1,2} = 2$. The eigenspace is

$$V_{\lambda_{1,2}} = \text{Null} \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

Thus, a basis for $V_{\lambda_{1,2}}$ is $\vec{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Note that A has only one linearly independent eigenvector. Thus, there exists no basis of $\mathbb{R}^2$ consisting of eigenvectors of A and therefore A is not a diagonalizable matrix.

(c)

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution: The characteristic polynomial of A is given by

$$p_A(\lambda) = \det(\lambda I_2 - A) = \begin{vmatrix} \lambda - 2 & -1 \\ -1 & \lambda \end{vmatrix} = \lambda(\lambda - 2) - 1 = \lambda^2 - 2\lambda - 1$$

Equating to zero, the eigenvalues are given by $\lambda_{1,2} = 1 \pm \sqrt{2}$. We now find a basis for each of the eigenspaces,

$$\begin{aligned} V_{\lambda_1} &= \text{Null}(\lambda_1 I_2 - A) = \text{Null} \begin{pmatrix} \lambda_1 - 2 & -1 \\ -1 & \lambda_1 \end{pmatrix} = \text{Null} \begin{pmatrix} -1 + \sqrt{2} & -1 \\ -1 & 1 + \sqrt{2} \end{pmatrix} \\ &\stackrel{\text{the rows are dependent}}{=} \text{Null} \begin{pmatrix} -1 + \sqrt{2} & -1 \\ 0 & 0 \end{pmatrix} = \text{Span} \left\{ \vec{v}_1 = \begin{pmatrix} 1 + \sqrt{2} \\ 1 \end{pmatrix} \right\} \end{aligned}$$

and similarly,

$$\begin{aligned} V_{\lambda_2} &= \text{Null}(\lambda_2 I_2 - A) = \text{Null} \begin{pmatrix} \lambda_2 - 2 & -1 \\ -1 & \lambda_2 \end{pmatrix} = \text{Null} \begin{pmatrix} -1 - \sqrt{2} & -1 \\ -1 & 1 - \sqrt{2} \end{pmatrix} \\ &\stackrel{\text{the rows are dependent}}{=} \text{Null} \begin{pmatrix} -1 - \sqrt{2} & -1 \\ 0 & 0 \end{pmatrix} = \text{Span} \left\{ \vec{v}_2 = \begin{pmatrix} 1 - \sqrt{2} \\ 1 \end{pmatrix} \right\} \end{aligned}$$

Clearly, $\vec{v}_1, \vec{v}_2$ are linearly independent. It follows that A is diagonalizable and that

$$P^{-1}AP = \text{diag}(\lambda_1, \lambda_2)$$

$$\text{where } P = \begin{pmatrix} \vec{v}_1 & \vec{v}_2 \end{pmatrix} = \begin{pmatrix} 1 + \sqrt{2} & 1 - \sqrt{2} \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix}$$

(d)

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 0 & -1 & 0 \\ 2 & 1 & 0 \end{pmatrix}$$

Hint. compute the characteristic polynomial by developing the determinant according to the second row.

Solution:

The characteristic polynomial of A is given by

$$\begin{aligned} p_A(\lambda) &= \det(\lambda I_3 - A) = \det \begin{pmatrix} \lambda - 3 & -1 & 1 \\ 0 & 1 + \lambda & 0 \\ -2 & -1 & \lambda \end{pmatrix} \stackrel{\text{second row}}{=} (\lambda + 1) \det \begin{pmatrix} \lambda - 3 & 1 \\ -2 & \lambda \end{pmatrix} \\ &= (\lambda + 1)(\lambda^2 - 3\lambda + 2) = (\lambda + 1)(\lambda - 1)(\lambda - 2) \end{aligned}$$

Therefore A has three distinct eigenvalues, $\lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 2$. We now compute

$$V_{\lambda_1} = \text{Null}(-I_3 - A) = \text{Null} \begin{pmatrix} -4 & -1 & 1 \\ 0 & 0 & 0 \\ -2 & -1 & -1 \end{pmatrix} \stackrel{R_1 \rightarrow R_1 - 2R_3}{=} \text{Null} \begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 0 \\ 2 & 1 & 1 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} \right\}$$

Similarly,

$$V_{\lambda_2} = \text{Null}(I - A) = \text{Null} \begin{pmatrix} -2 & -1 & 1 \\ 0 & 2 & 0 \\ -2 & -1 & 1 \end{pmatrix} \stackrel{R_3 \rightarrow R_3 - R_1}{=} \text{Null} \begin{pmatrix} -2 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\}$$

and

$$V_{\lambda_3} = \text{Null}(2I - A) = \text{Null} \begin{pmatrix} -1 & -1 & 1 \\ 0 & 3 & 0 \\ -2 & -1 & 2 \end{pmatrix} \stackrel{R_3 \rightarrow R_3 - 2R_1}{=} \text{Null} \begin{pmatrix} -1 & -1 & 1 \\ 0 & 3 & 0 \\ 0 & -1 & 0 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

We now define $P = (\vec{v}_1 \quad \vec{v}_2 \quad \vec{v}_3)$. We compute

$$\det(P) = \det \begin{pmatrix} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 \end{pmatrix} = \det \begin{pmatrix} 1 & 1 & 1 \\ -3 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix} \stackrel{\text{direct formula}}{=} -3$$

and therefore P is invertible, which implies that its columns are a basis $\mathbb{R}^3$. Thus, there is a basis consisting of eigenvectors of A and therefore A is diagonalizable and $P^{-1}AP = \text{diag}(-1, 1, 2)$.

2. Let $A = \begin{pmatrix} 2 & 1 \\ -6 & 7 \end{pmatrix}$. Compute A^{100} .

Solution:

First, we compute the characteristic polynomial of A :

$$\begin{aligned} p_A(\lambda) &= \det(\lambda \cdot I_n - A) = \det \begin{pmatrix} \lambda - 2 & -1 \\ 6 & \lambda - 7 \end{pmatrix} \\ &= (\lambda - 2)(\lambda - 7) + 6 = \lambda^2 - 9\lambda + 20 = (\lambda - 4)(\lambda - 5). \end{aligned}$$

We know the eigenvalues of A are exactly the roots of $p_A(\lambda)$. Thus $p_A(\lambda) = 0 \Leftrightarrow \lambda_{1,2} = 4, 5$.

For each eigenvalue we now find a corresponding eigenvector:

$\lambda_1 = 4$:

$$V_{\lambda_1} = \text{Null}(\lambda_1 \cdot I_n - A) = \text{Null}\begin{pmatrix} 2 & -1 \\ 6 & -3 \end{pmatrix}$$

$$\stackrel{R_2 \rightarrow R_2 - 3R_1}{=} \text{Null}\begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} = \left\{ \begin{pmatrix} t \\ 2t \end{pmatrix} : t \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}.$$

Thus, $\vec{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is an eigenvector which corresponds to $\lambda_1 = 4$.

$\lambda_2 = 5$:

$$V_{\lambda_2} = \text{Null}(\lambda_2 \cdot I_n - A) = \text{Null}\begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix}$$

$$\stackrel{R_2 \rightarrow R_2 - 2R_1}{=} \text{Null}\begin{pmatrix} 3 & -1 \\ 0 & 0 \end{pmatrix} = \left\{ \begin{pmatrix} t \\ 3t \end{pmatrix} : t \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right\}.$$

Thus, $\vec{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ is an eigenvector which corresponds to $\lambda_2 = 5$.

We now denote

$$P = \begin{pmatrix} \vec{v}_1 & \vec{v}_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}.$$

Note that $\det P = 3 - 2 = 1 \neq 0$ which implies that P is invertible. We know that

$$P^{-1}AP = \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}$$

Note that $P^{-1} = \frac{\text{adj}(P)}{\det P} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$. It follows that

$$\begin{aligned} A^{100} &= P \cdot \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}^{100} \cdot P^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}^{100} \cdot \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 4^{100} & 0 \\ 0 & 5^{100} \end{pmatrix} \cdot \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \cdot 4^{100} & -4^{100} \\ -2 \cdot 5^{100} & 5^{100} \end{pmatrix} \\ &= \begin{pmatrix} 3 \cdot 4^{100} - 2 \cdot 5^{100} & -4^{100} + 5^{100} \\ 6 \cdot 4^{100} - 6 \cdot 5^{100} & -2 \cdot 4^{100} + 3 \cdot 5^{100} \end{pmatrix}. \end{aligned}$$

3. Note: there is no logical connection between the items of this question.

- (a) i. Find a matrix $A \in M_3(\mathbb{Z}_5)$ for which $p_A(\lambda) = 1 + 3\lambda + \lambda^2 + \lambda^3$.

Hint. Use HW12Q6 in LA1.

Solution: Consider the matrix defined by

$$A = \begin{pmatrix} 0 & 0 & 4 \\ 1 & 0 & 2 \\ 0 & 1 & 4 \end{pmatrix}$$

We compute

$$\begin{aligned} p_A(\lambda) &= \det(\lambda I_3 - A) = \det \begin{pmatrix} \lambda & 0 & -4 \\ -1 & \lambda & -2 \\ 0 & -1 & \lambda - 4 \end{pmatrix} = \det \begin{pmatrix} \lambda & 0 & 1 \\ 4 & \lambda & 3 \\ 0 & 4 & \lambda + 1 \end{pmatrix} \\ &= \lambda^2(\lambda + 1) + 4^2 - 3 \cdot 4 \cdot \lambda = \lambda^3 + \lambda^2 + 3\lambda + 1 \end{aligned}$$

ii. Prove that A is not diagonalizable in $M_3(\mathbb{Z}_5)$.

Solution: We prove that A has no eigenvalues in $\mathbb{Z}_5$, and therefore A is not diagonalizable. As $p_A(\lambda) = 1 + 3\lambda + \lambda^2 + \lambda^3$ then we can compute all of the values of $p_A(\lambda)$ for every $\lambda \in \mathbb{Z}_5$,

$$p_A(0) = p_A(1) = 1, p_A(2) = 4, p_A(3) = 1, p_A(4) = p_A(-1) = 1 - 3 + 1 - 1 = 3$$

Thus $p_A(\lambda) \neq 0$ for every $\lambda \in \mathbb{Z}_5$ and therefore A has no eigenvalues in $\mathbb{Z}_5$.

(b) Let $A(x) \in M_n(F_2[x])$. Prove that $\det(A(x)) \in F_{2n}[x]$.

Hint. follow a similar theorem that was studied in class.

Solution: We prove this claim by induction.

Base case: ($n=1$) In this case, there exist $a, b, c \in F$ such that $A(x) = a + bx + cx^2$. Thus, $\det(A(x)) = a + bx + cx^2 \in F_1[x]$ and the claim holds.

Induction step: ($n-1 \Rightarrow n$). Let $2 \leq n \in \mathbb{N}$. Developing the determinant of $A(x)$ according to the first column, we obtain

$$\det(A(x)) = \sum_{i=1}^n (-1)^{i+1} [A(x)]_{i1} M_{i1}^{(A(x))}$$

By the induction hypothesis, for every $1 \leq i \leq n$ we have $M_{i1}^{(A(x))} \in F_{2n-2}[x]$. As $[A(x)]_{i1} \in F_2[x]$ then for every $1 \leq i \leq n$ we have $[A(x)]_{i1} M_{i1}^{(A(x))} \in F_{2n}[x]$. As $F_n[x]$ is closed under linear combinations, we conclude that $\det(A(x)) \in F_{2n}[x]$.

4. Student X had a good day and went over HW5Q3 again. If A is invertible, then AB and BA are similar. It means they must have the same eigenvalues. But what if A is not invertible? she also recalled that it is possible for AB and AB to be defined even when A, B are not square matrices. In that case, the condition that A is invertible isn't even relevant. Is there a situation where AB and BA have different eigenvalues?

Let $A \in M_{m \times n}(F)$ and let $B \in M_{n \times m}(F)$.

(a) Let $0_F \neq \lambda \in F$ and let $\vec{0} \neq \vec{v} \in F^m$. Suppose that $(\lambda, \vec{v})$ is an eigenpair of AB . Prove that $(\lambda, B\vec{v})$ is an eigenpair of BA .

Hint. explain why $B\vec{v} \neq \vec{0}_n$.

Solution:

As $(\lambda, \vec{v})$ is an eigenpair of AB then $AB\vec{v} = \lambda\vec{v}$. We first show that $B\vec{v} \neq \vec{0}_n$. ATC that $B\vec{v} = \vec{0}_n$. Then

$$\lambda\vec{v} = AB\vec{v} = A \cdot \vec{0}_n = \vec{0}_m$$

As $\lambda \neq 0_F$ then $\vec{v} = \vec{0}_m$, in contradiction to our assumption on $\vec{v}$. We now note that

$$BA \cdot B\vec{v} = B \cdot \lambda\vec{v} = \lambda \cdot B\vec{v}$$

and therefore $B\vec{v}$ is an eigenvector of BA .

- (b) Conclude that if $n = m$ then AB and BA have exactly the same eigenvalues, that is, prove that for every $\lambda \in F$, λ is an eigenvalue of AB if and only if λ is an eigenvalue of BA .

Hint. Consider the two cases $\lambda \neq 0_F$ and $\lambda = 0_F$ separately.

Solution: Let $\lambda \in F$ and suppose that λ is an eigenvalue of AB . If $\lambda \neq 0_F$ then we already know from item (a) that λ is an eigenvalue of BA . If $\lambda = 0_F$, then by HW6Q5(a), AB is not invertible. It follows that $\det(BA) = \det(AB) = 0_F$ and therefore BA is not invertible. Thus, by HW6Q5(a), 0_F is an eigenvalue of BA .

The reverse direction is similar, by replacing A and B .

5. Student X drifted in thought while watching Netflix. A matrix $A \in M_n(F)$ could be multiplied by column vectors as long as they appear on the right side of A and the result is a column vector. It could be an eigenvector or not, and when A is diagonalizable, these eigenvectors appear in the columns of P (at least if there are enough to form a basis). But... the matrix A could be multiplied by a row vector from the left: If $\vec{v} \in M_{1 \times n}(F)$ and if $A \in M_n(F)$ then $\vec{v} \cdot A \in M_{1 \times n}(F)$. Many questions started popping up: could there be left eigenvectors? Are there also left eigenvalues? Is the government hiding them from us? She decided to ask her TA for help.

Let $A \in M_n(F)$ and let $\lambda \in F$.

- (a) Prove that λ is an eigenvalue of A if and only if it is an eigenvalue of A^t .

Hint. Prove that $p_{A^t}(\lambda) = 0_F$.

Solution: We have

$$p_{A^t}(\lambda) = \det(A^t - \lambda I_n) = \det[(A - \lambda I_n)^t] = \det(A - \lambda I_n) = p_A(\lambda) = 0$$

and therefore λ is an eigenvalue of A^t .

- (b) Let $\vec{0} \neq \vec{v} \in F^n$. Prove that $(\vec{v})^t$ is a left eigenvector of A with the corresponding eigenvalue λ if and only if $\vec{v}$ is a right eigenvector of A^t , corresponding to the same eigenvalue. Conclude that the left and the right eigenvalues of A are the same.

Solution: By definition, $(\vec{v})^t$ is a left eigenvector of A with the corresponding eigenvalue λ if and only if $(\vec{v})^t \cdot A = \lambda \cdot (\vec{v})^t$. Transposing both sides of the equation, we get the equivalent equation

$$A^t \cdot \vec{v} = \lambda \cdot \vec{v}$$

and this is if and only if $\vec{v}$ is a right eigenvector of A^t , corresponding to the eigenvalue λ .

- (c) Suppose that A is diagonalizable and let $P \in M_n(F)$ be an invertible matrix such that $P^{-1}AP$ is diagonal. Conclude that the rows of P^{-1} are a basis consisting of left eigenvectors of A .

Solution: As $P^{-1}AP$ is diagonal, then there exist $\lambda_1, \dots, \lambda_n \in F$ such that $P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n)$. Transposing both sides of the equation, we get

$$P^t A^t (P^t)^{-1} = (\text{diag}(\lambda_1, \dots, \lambda_n))^t = \text{diag}(\lambda_1, \dots, \lambda_n)$$

Thus, A^t is diagonalizable, and the eigenvectors of A^t are the columns of P^t , which are the rows of P . By item (b), the row of P are left eigenvectors of A .

6. Let $A, B \in M_n(F)$ and let $\lambda \in F$. Prove or disprove each of the following statements:

- (a) A is singular if and only if 0_F is an eigenvalue of A .

Solution:

True. We know that 0_F is an eigenvalue of A if and only if

$$(-1)^n \cdot \det(A) = \det(-A) = \det(0 \cdot I_n - A) = p_A(0_F) = 0_F$$

and this holds if and only if $\det(A) = 0_F$, i.e., if and only if A is singular.

Alternative solution:

We know that A is singular if and only if there exists an $\vec{0} \neq \vec{x} \in F^n$ such that $A\vec{x} = \vec{0} = 0_F \cdot \vec{x}$, i.e, if and only if 0_F is an eigenvalue of A (and the corresponding eigenvector is $\vec{x}$).

- (b) Suppose that λ is an eigenvalue of A . Then λ^2 is an eigenvalue of A^2 .

Solution:

True. As λ is an eigenvalue of A then there exists an eigenvector $\vec{0} \neq \vec{v} \in F^n$ which corresponds to λ , that is, such that $A\vec{v} = \lambda\vec{v}$. It follows that

$$A^2\vec{v} = A \cdot A\vec{v} = A \cdot \lambda\vec{v} = \lambda A\vec{v} = \lambda \cdot \lambda\vec{v} = \lambda^2 \cdot \vec{v}$$

and therefore, λ^2 is an eigenvalue of A^2 , and its corresponding eigenvector is $\vec{v}$.

Alternative solution: As λ is an eigenvalue of A , then $p_A(\lambda) = 0_F$. Thus

$$\begin{aligned} p_{A^2}(\lambda) &= \det(\lambda^2 I_n - A^2) = \det[(\lambda I_n - A)(\lambda I_n + A)] \\ &= \det(\lambda I_n - A) \det(\lambda I_n + A) = 0_F \cdot \det(\lambda I_n + A) = 0_F \end{aligned}$$

and therefore, λ^2 is an eigenvalue of A^2 .

- (c) Suppose that λ^2 is an eigenvalue of A^2 . Then either λ is an eigenvalue of A , or $-\lambda$ is an eigenvalue of A .

Solution:

True. As λ^2 is an eigenvalue of A^2 , then $p_{A^2}(\lambda^2) = 0_F$. We develop

$$p_{A^2}(\lambda^2) = \det(\lambda^2 \cdot I_n - A^2) = \det[(\lambda I_n - A)(\lambda I_n + A)] = \det(\lambda I_n - A) \det(\lambda I_n + A)$$

and as $p_{A^2}(\lambda^2) = 0_F$ then $\det(\lambda I_n - A) = 0_F$ or $\det(\lambda I_n + A) = 0_F$. If $\det(\lambda I_n - A) = 0_F$ then $p_A(\lambda) = 0_F$ and therefore λ is an eigenvalue of A . Otherwise, $\det(\lambda I_n + A) = 0_F$ and therefore

$$p_A(-\lambda) = \det(-\lambda I_n - A) = -(1)^n \cdot \det(\lambda I_n + A) = 0_F$$

and therefore $-\lambda$ is an eigenvalue of A .

- (d) If A, B are diagonalizable, then $A + B$ is diagonalizable.

Solution: False. We choose $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$. Note that as A, B are upper triangular then their eigenvalues lie on their diagonal. The eigenvalues of A are 1, 0. The eigenvalues of B are -1, 0. Thus both matrices have two distinct eigenvalues and therefore they are diagonalizable. Note that

$$A + B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

We now prove that $A + B$ is not diagonalizable. ATC that $A + B$ is diagonalizable. As $A + B$ is upper triangular then its eigenvalues lie on its diagonal and therefore $A + B$ has a single eigenvalue, namely $\lambda = 0$. As $A + B$ is diagonalizable, then it is similar to the zero matrix. Contradiction, as the only matrix that is similar to the zero matrix is the zero matrix.

- (e) i. Suppose $\lambda = 1_F$ is the only eigenvalue of A . If $A^2 = I_n$ then $A = I_n$.

Solution: False. Choose

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{Z}_2).$$

Note that $A^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$. Note also that the characteristic polynomial of A is

$$p_A(\lambda) = \det(\lambda \cdot I_2 - A) = \det \begin{pmatrix} \lambda & -1 \\ -1 & \lambda \end{pmatrix} = \lambda^2 - 1$$

We know that $\lambda \in \mathbb{Z}_2$ is an eigenvalue of A iff $p_A(\lambda) = 0$, that is, if and only if $\lambda^2 = 1 \Leftrightarrow \lambda = 1$. But $A \neq I_2$.

ii. Suppose $1_F \neq -1_F$ and that $\lambda = 1$ is the only eigenvalue of A . If $A^2 = I_n$ then $A = I_n$.

Solution:

True. It is sufficient to show that for every $\vec{v} \in F^n$ we have $A \cdot \vec{v} = \vec{v}$.

Let $\vec{v} \in F^n$ and denote $\vec{w} = A \cdot \vec{v} - \vec{v} \in F^n$. We want to show $\vec{w} = \vec{0}$. ATC that $\vec{w} \neq \vec{0}$. Then

$$\begin{aligned} A \cdot \vec{w} &= A \cdot (A \cdot \vec{v} - \vec{v}) = A^2 \cdot \vec{v} - A \cdot \vec{v} = I_n \cdot \vec{v} - A \cdot \vec{v} \\ &= \vec{v} - A \cdot \vec{v} = -(A \cdot \vec{v} - \vec{v}) = -\vec{w} = -1_F \cdot \vec{w}. \end{aligned}$$

As $\vec{w} \neq \vec{0}$, then $\lambda = -1_F$ is an eigenvalue of A and $\vec{w}$ is a corresponding eigenvector. Since we assumed $1_F \neq -1_F$, we get a contradiction to the assumption that $\lambda = 1_F$ is the only eigenvalue of A .